

Logical Energy as Test-Time Compute: Scoring and Repairing Neural Outputs with a Differentiable Theorem-Proving Energy-Based Model

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Technical Report · Project ThermoLogic · Reproducible PyTorch prototype

Scope & honesty note. This is an engineering technical report / proof-of-concept, not a peer-reviewed claim of a new paradigm. It builds directly on established neuro-symbolic work (Logic Tensor Networks, DeepProbLog, Semantic Loss, Neural Theorem Provers, and analyses of differentiable fuzzy operators). Every number is produced by the accompanying open code with fixed seeds. The contribution is (i) a clean, fully-reproducible implementation, and (ii) two small original observations about *test-time energy repair* and how *reasoning depth maps to test-time compute*.

Abstract. Neural models trained by maximum likelihood have no internal mechanism forbidding logically inconsistent outputs. We present **Project ThermoLogic**, a compact PyTorch system that fuses Differentiable Theorem Proving (relaxing discrete rules into fuzzy-logic t-norms) with an Energy-Based Model: a belief state's energy is ~ 0 when it satisfies every rule and grows exponentially as rules are violated. A small MLP proposes probabilistic truth values; the energy acts as a differentiable prover. On an 11-atom benchmark with a genuine world-level train/test split we report three findings. (1) A plain supervised network matches ThermoLogic on raw accuracy (0.955 vs 0.970) but its outputs are logically inconsistent (mean energy 0.95); the energy layer's value is a **label-free consistency guarantee** (energy 0.001), not accuracy. (2) **Test-time energy repair** — gradient descent on the energy — turns an inconsistent output on unseen inputs into a valid one (0.935 $\rightarrow$ 0.971 accuracy; energy 0.90 $\rightarrow$ 0.0006). (3) In a proof chain, truth propagates **one hop at a time**: inference cost grows linearly with reasoning depth, and the t-norm sets the propagation speed (~ 3.4 descent steps/hop for Łukasiewicz vs ~ 1.0 for Gödel/Product). The mechanism ships as a small library that scores and repairs any model's structured output against hard rules at a controllable compute budget.

Keywords: neuro-symbolic AI, energy-based models, differentiable logic, t-norms, test-time compute, constraint satisfaction, guardrails, PyTorch.

1. Introduction

A maximum-likelihood objective rewards fitting the data distribution; it never assigns infinite cost to a self-contradictory output. Symbolic systems are the opposite — rigorously consistent but non-differentiable. Project ThermoLogic asks a narrow, testable question: *can a neural network's continuous, probabilistic beliefs be constrained — and repaired — by discrete logical rules, using only backpropagation?* The organizing metaphor is thermodynamic: logical validity corresponds to low energy (a ground state), a contradiction to high energy. Because the energy is differentiable, it can be minimized at training time (as a loss) and at test time (as an inference procedure that repairs a given output).

Contributions. (1) A clean, reproducible DTP + EBM implementation with three t-norm families and residuated implications. (2) A label-free consistency signal + repair operator, and the finding that a plain supervised net matches on accuracy but not consistency — so the value is reliability, not accuracy (§5.3). (3) Test-time energy repair framed as inference-time compute, with a measured compute–consistency curve (§5.2). (4) The truth-propagation wave: reasoning depth $\mapsto$ test-time compute is linear, and the t-norm sets the speed (§5.1). (5) A small product — `score()` / `repair()` — and a worked config-validation use case.

2. Method

Throughout, an **atom** is a single atomic proposition — one true/false statement such as `Rain` or `sso_enabled` (the standard logic term for the smallest unit of a rule; unrelated to physics atoms despite the thermodynamic framing).

Let $p_i \in [0,1]$ be the belief that atom i is true. A literal's truth is p_i (or $1-p_i$ if negated). A Horn rule $L_1 \wedge \dots \wedge L_k \rightarrow C$ scores as $\text{sat} = \text{residuum}(\otimes_i \text{truth}(L_i), \text{truth}(C))$. Modus ponens is emergent: with the antecedent believed, $\text{residuum}(1, p_C) = p_C$, so $\text{sat} \rightarrow 1$ requires $p_C \rightarrow 1$. Per-rule energy is $E_r = \exp(\beta(1-\text{sat}_r)) - 1$; total $E = \text{mean}_r E_r$. Valid $\Rightarrow E=0$; contradictions grow exponentially with inverse temperature β .

Table 1 — Operator mapping per t-norm family.

Discrete	Product	Łukasiewicz (default)	Gödel
$\neg a$	$1-a$	$1-a$	$1-a$
$a \wedge b$	$a \cdot b$	$\max(0, a+b-1)$	$\min(a,b)$
$a \Rightarrow c$ (residuum)	1 if $a \leq c$ else c/a	$\min(1, 1-a+c)$	1 if $a \leq c$ else c

Test-time repair (the key operator). Given any belief state p (e.g. a model's output on a novel input), hold trusted atoms fixed and run gradient descent on E over the rest, in logit space: $p^* = \text{argmin}_p E(p)$. Because minimizing E enforces every rule, this repairs an inconsistent output to the nearest valid state; the number of steps is a compute budget.

3. Benchmark

The expanded knowledge base has 11 atoms and 9 rules, yielding 18 distinct logically-consistent worlds (294 of 2048 full Boolean worlds are consistent). Five atoms are observed causes (Rain, Sprinkler, Cold, HeaterOn, Windy); six are derived (Cloudy, WetGround, Slippery, Ice, Warm, Chilly) via rules including two-literal antecedents ($\text{Cold} \wedge \text{WetGround} \rightarrow \text{Ice}$) and a two-hop chain ($\text{Rain} \rightarrow \text{WetGround} \rightarrow \text{Slippery}$), plus negative constraints ($\text{Rain} \rightarrow \neg \text{Sprinkler}$). We hold out entire worlds: train on 12, test on 6 never seen. Inputs are noisy encodings of the causes ($\sigma=0.12$) with 3 distractor dimensions.

4. Experiments

4.1 Logical energy as test-time compute (the propagation wave)

Starting from a state where a cause is true but a chain of derived atoms is false, test-time energy descent (plain SGD) turns atoms on one hop at a time — atom depth d activates only after $d-1$. Activation step is a strictly monotone, near-linear function of depth; the t-norm sets the slope (Łukasiewicz 3.4 steps/hop; Product/Gödel ~ 1.0). The clean monotone wave requires momentum-free SGD; the robust claim is the within-chain wave and linear depth-cost at fixed length.

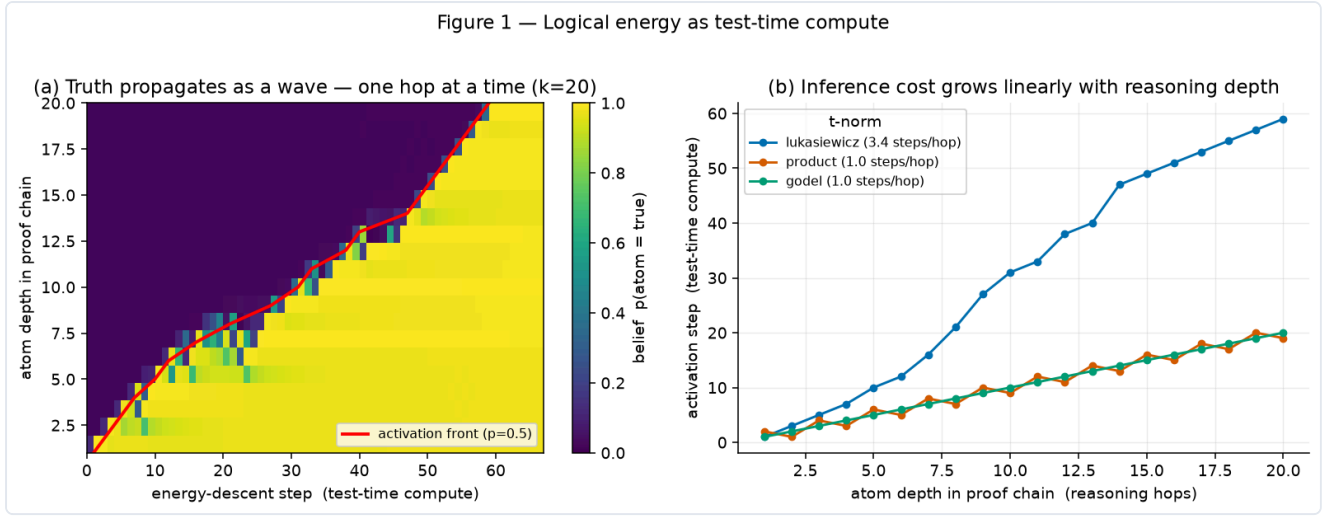


Figure 1. (a) Truth propagates as a wave — one hop at a time. (b) Inference cost grows linearly with reasoning depth; the t -norm sets the propagation speed.

4.2 Generalization and test-time repair

On worlds never seen in training, a single feed-forward pass generalizes only partially and leaves outputs logically inconsistent (high energy). Repair fixes them, and is a compute knob: more steps $\rightarrow$ lower energy and higher accuracy, plateauing near 50 steps.

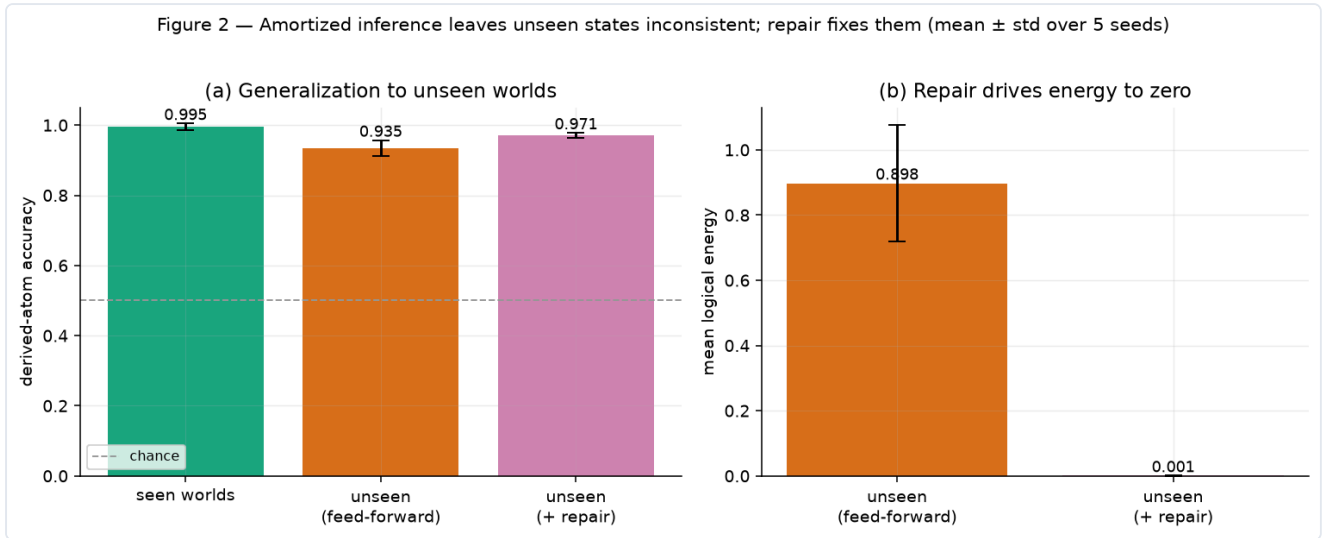


Figure 2. Amortized inference leaves unseen states inconsistent; repair drives energy to zero (mean $\pm$ std, 5 seeds).

Table 2 — Generalization (mean $\pm$ std, 5 seeds).

Split	Derived accuracy	Mean energy
Seen worlds	0.995 ± 0.010	~ 0
Unseen (feed-forward)	0.935 ± 0.022	0.898 ± 0.178
Unseen (+ repair)	0.971 ± 0.007	0.001 ± 0.000
Cause recovery (unseen)	0.981 ± 0.004	—

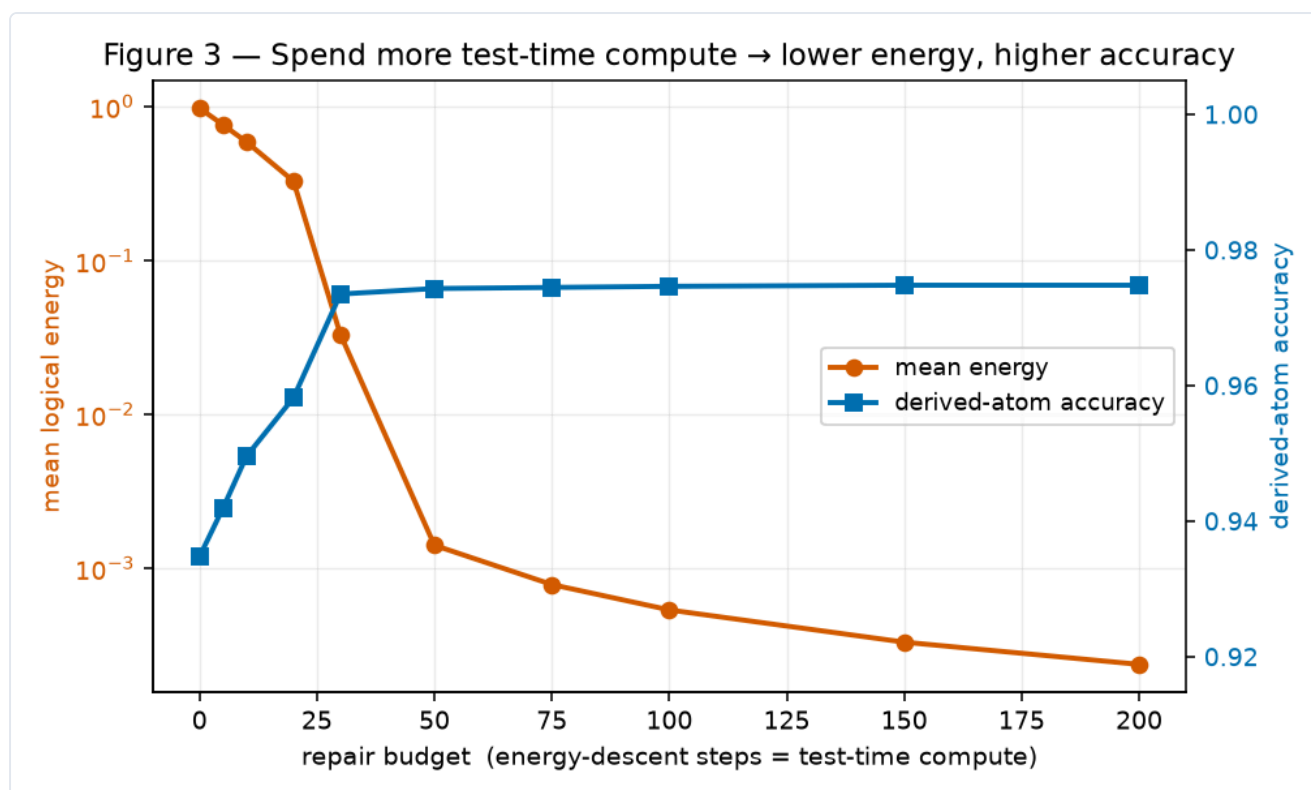


Figure 3. Test-time compute vs. energy and accuracy: spend more descent steps → lower energy, higher accuracy.

4.3 Baselines — the value is consistency, not accuracy

A plain supervised network (BCE on all atoms, no logic) reaches 0.955 accuracy on unseen worlds — comparable to ThermoLogic + repair (0.970). But its outputs are logically inconsistent (energy 0.946), with no signal that anything is wrong. Only the energy provides a label-free consistency guarantee (0.001). **This is the honest headline: ThermoLogic is a reliability mechanism, not a way to beat a classifier on accuracy.**

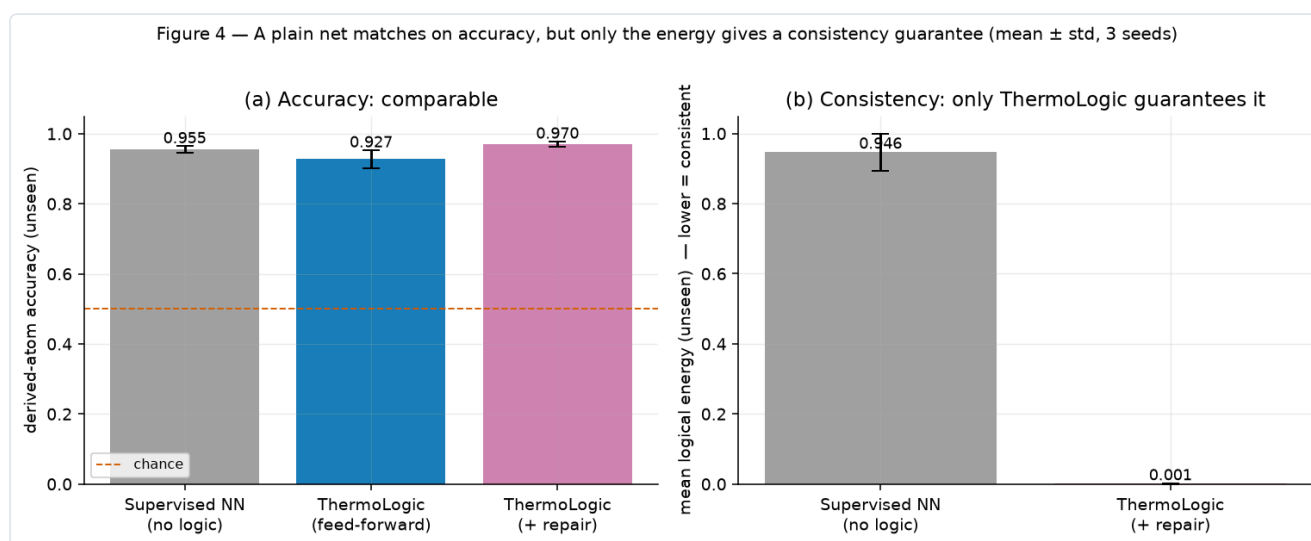


Figure 4. Comparable accuracy; only the energy guarantees consistency (mean ± std, 3 seeds).

Table 3 — Baselines on unseen worlds (mean $\pm$ std, 3 seeds).

Method	Derived accuracy	Mean energy (consistency)
Supervised NN (no logic)	0.955 ± 0.010	$0.946 \pm 0.053 \times$ inconsistent
ThermoLogic (feed-forward)	0.927 ± 0.026	0.90
ThermoLogic (+ repair)	0.970 ± 0.008	$0.001 \pm 0.000 \checkmark$

4.4 Robustness and t-norm comparison

Repair helps most when causes are recoverable (low–moderate noise); at high noise ($\sigma \geq 0.3$) mis-recovered causes cap accuracy. As more worlds are held out, generalization degrades gracefully. Among t-norms, Łukasiewicz's smoother gradients give the best accuracy and repairability (0.970 vs 0.876 Product, 0.878 Gödel) — the default.

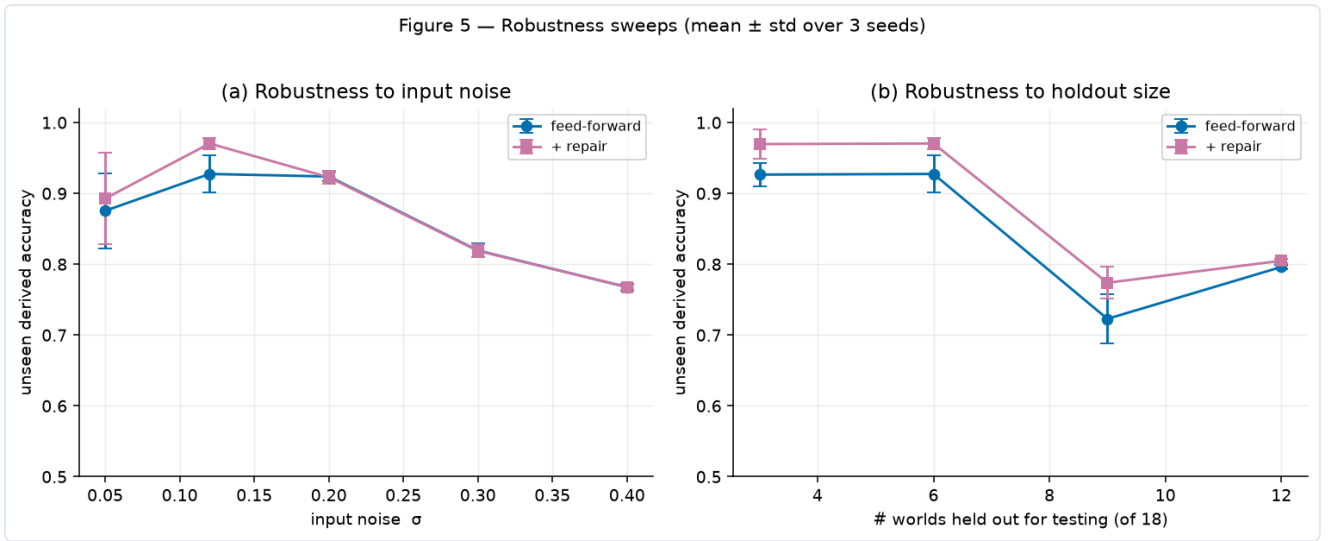


Figure 5. Robustness to input noise (a) and holdout size (b); mean $\pm$ std, 3 seeds.

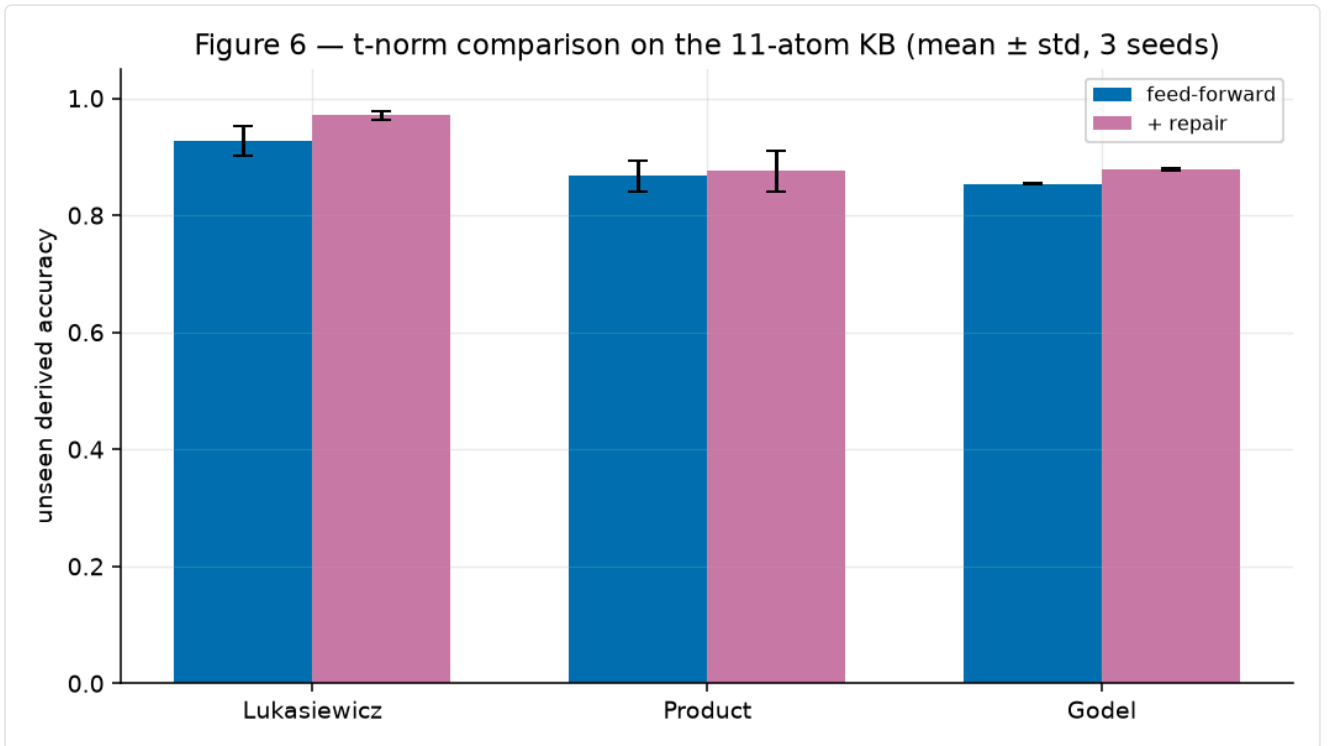


Figure 6. t-norm comparison on the 11-atom KB.

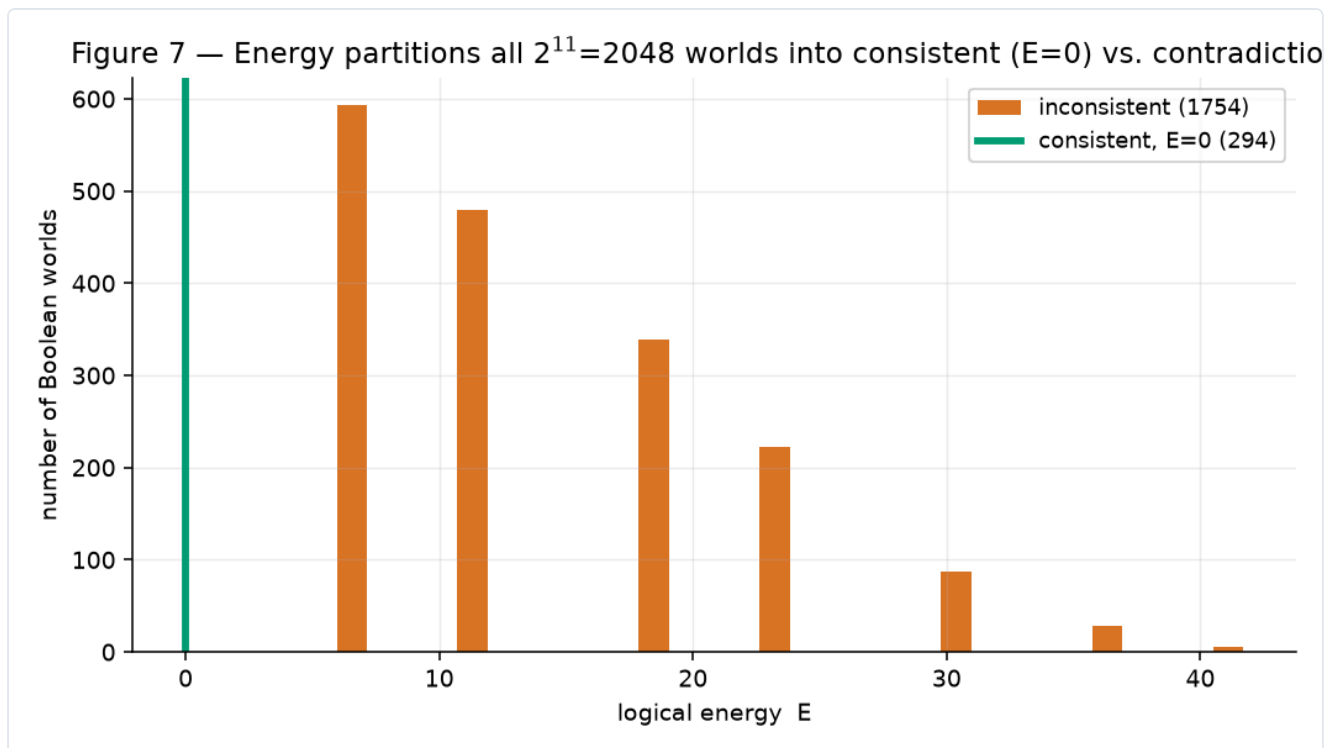


Figure 7. Energy partitions all $2^{11} = 2048$ Boolean worlds into consistent ($E=0$) vs. contradictions (294 consistent; the rest carry energy in $[5.96, 41.69]$).

5. The product

The mechanism ships as `thermologic`, a differentiable logic-energy layer that drops after any model: `score()` gives a label-free inconsistency signal, `violations()` names the broken rules, and `repair(budget=...)` returns the nearest valid output at a controllable compute budget. A worked SaaS config-validation example flags an invalid plan (e.g. a Pro request for enterprise-only SSO), names the rule, and repairs to the nearest valid configuration. v1 scope: structured/tabular outputs with propositional rules; general LLM-text validation is roadmap, not a claim.

6. Discussion & limitations

The honest value proposition is a label-free consistency guarantee and repair, not beating supervised learning on accuracy. The energy adds an iterative, budgetable inference step — a small instance of test-time compute for logical consistency. Limitations: 11 atoms, propositional Horn-style rules, synthetic data; no first-order variables, no defeasible rules, no real LLM. Soft energy makes contradictions expensive, not impossible (a crisp projection step would give a hard guarantee). The clean propagation wave needs momentum-free SGD, and mean-energy normalization couples wave speed to chain length.

7. Conclusion

A differentiable fuzzy-logic energy lets a neural network's beliefs be scored and repaired against hard rules by backpropagation alone. Empirically the value is reliability: comparable accuracy to a plain net but with a label-free consistency guarantee, an inference-time repair that scales with a compute budget, and a clean picture of reasoning depth as test-time compute — a small, honest, reproducible substrate for constraint-aware ML.

References

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References are indicative of the intellectual context; this is a self-contained prototype, not a comparative benchmark against these systems. Code, figures, and this report regenerate from the open repository with fixed seeds.